
Theorem Evolution Graphs: A Structured Representation of How Mathematical Theorems Build on One Another

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Abstract

Much of mathematics research can now be automated: proofs can be automatically generated, formalized, verified, and written up for dissemination. Coming up with a promising conjecture is harder. Automated methods can generate conjectures, but figuring out whether one is sound, plausible, and interesting to work on still takes considerable human oversight, since a conjecture cannot be verified the way a proof can. Mathematicians might typically do this by surveying prior results and identifying which ideas can be meaningfully extended. Inspired by this evolutionary process, in which new results build on existing ones, we introduce the theorem evolution graph (TEG), a graph where nodes are theorems and edges are typed relations naming how one theorem builds on another (e.g. generalizing it, weakening a hypothesis). We construct TEGs over 1,213 arXiv papers, seeded both from open mathematical conjecture datasets and from iterative citation search from a seminal paper. We prompt frontier LLMs to extend a given theorem augmented with different contexts, and judge the similarity of resulting conjectures to follow-up work that actually cited it. TEG-augmented prompting outperforms both a citation-graph baseline and zero-shot prompting (12.5% vs. 9.8% and 8.2%). These results demonstrate the effectiveness of encoding the evolutionary structure of mathematical progress for automated conjecture generation.

1 Introduction

AI systems are increasingly capable of contributing to mathematical research where outputs can be mechanically verified, including theorem formalization and proving [Yang et al., 2023, Dong and Ma, 2025]. These systems typically begin with a theorem, conjecture, or formal goal.

Generating a promising conjecture cannot be as easily verified. Candidates can be checked for counterexamples [Chuharski et al., 2024], generated in a formal language to filter malformed or trivial statements [Onda et al., 2025], or coupled with proving to provide a signal of tractability [Dong and Ma, 2025]. But whether a conjecture is plausible, novel, and interesting enough to pursue is subjective. In practice, a mathematician might build on prior work: weakening an assumption, strengthening a conclusion, generalizing to a broader class of objects, or transferring a proof technique to a new setting [Pólya, 1990, Kitcher, 1984, Corfield, 2003, Thurston, 2006]. These extensions link theorems into lines of development. Existing graph representations miss this structure: citation graphs connect papers, while knowledge graphs encode factual relations among entities.

We introduce the *Theorem Evolution Graph* (TEG), in which nodes are theorems and typed edges describe how one theorem develops from another. We construct TEGs with a two-stage LLM-based pipeline: (1) reduce, which extracts a canonical representation of a paper’s central theorem and its dependencies, and (2) compare, which identifies typed evolutionary relations between related

theorems (i.e. a generalization). We apply this pipeline to 1,213 arXiv papers, seeded from mathematical conjecture datasets [Brown, 2026, Dekoninck et al., 2026] and from seminal work on stability conditions in triangulated categories [Bridgeland, 2007].

We evaluate TEGs as context for conjecture generation by comparing generated conjectures against work that later built on the source theorem, using published follow-up work as a proxy for conjectures that mathematicians found worth pursuing. Under an ensemble of LLM judges, TEG-augmented prompting matches follow-up work 12.5% of the time, compared with 9.8% for citation-graph context and 8.2% for zero-shot prompting. This suggests that representing how mathematical results evolve provides useful signal for proposing what might come next.

In summary, we:

- introduce the Theorem Evolution Graph (TEG), a new representation of mathematical progress that encodes how theorems build on one another.
- develop a Reduce–Compare pipeline to construct TEGs over 1,213 arXiv papers.
- show that TEG context increases how often generated conjectures match published follow-up work, compared to citation-graph and zero-shot baselines (12.5% vs. 9.8% and 8.2%).

2 Related work

Automated conjecture generation. Automated conjecture-generation methods use signals such as counterexample search [Chuharski et al., 2024], formalization [Onda et al., 2025], and interaction with theorem provers [Dong and Ma, 2025, Poesia et al., 2024]. Other systems shift attention to problem selection: Zheng et al. [2026] retrieve open conjectures from the mathematical literature and recommend which should receive model and expert effort. Related work in scientific discovery similarly uses the literature to ground or guide candidate research directions [Wang et al., 2023, Vasu et al., 2025, Jansen, 2026]. Our work is complementary: rather than retrieving an existing problem, we use the structure of prior mathematical results to provide context for generating new conjectures.

Graph representations of scientific knowledge. Knowledge graphs represent typed relations among entities [Hogan et al., 2021], while temporal knowledge graphs model how these relations vary over time [Li et al., 2021, Cai et al., 2022]. Scientific literature is commonly represented with citation graphs, which support search and recommendation [Ammar et al., 2018, Cohan et al., 2020]; citation-intent methods additionally characterize why one paper cites another [Valenzuela et al., 2015]. TEGs instead represent the evolution of mathematical results themselves: nodes are theorems, and typed edges encode the ways in which later results build on earlier ones.

3 Constructing Theorem Evolution Graphs

We construct TEGs in two stages.

Reduce. Reduce takes the full $\text{\LaTeX}$ source of a paper and prompts gpt-5.4 (temperature 0) to extract the paper’s central theorem, along with other definitions, lemmas, propositions, and corollaries introduced in the paper. For example, reducing Helfgott [2008] returns its result about growth in $SL_2(\mathbb{Z}/p\mathbb{Z})$: roughly, repeatedly multiplying elements of a generating set either causes the set to grow rapidly or shows that it is confined to a proper subgroup. Details are in Appendix A.2.

Compare. Compare identifies how one theorem builds on another. We first retrieve potentially related theorem pairs using text-embedding-3-small, retaining pairs above cosine similarity $\tau = 0.7$. We review prior accounts of mathematical discovery and progress [Isaacson, 1978, Pólya, 1990, Kitcher, 1984, Thurston, 2006, Corfield, 2003] to develop a taxonomy of recurring ways in which later theorems develop from earlier ones. The resulting relation types capture changes to a theorem’s statement, such as generalizing or strengthening it, as well as changes in how a result is proved, applied, or reformulated. The taxonomy is in Table 2. We then prompt gpt-5.4 with the two reduced theorems, including their statements, assumptions, and proof approaches, and ask whether the later theorem substantively develops from the earlier one. If so, the model assigns one of the 13 relations; otherwise, no edge is added. For example, comparing Helfgott’s result with Breuillard et al.

Table 1: The three paper collections from TEGs are constructed.

TEG	Seeds	Papers
BRIDGELAND	1	500
OPENCONJECTURE	50	427
ARXIVMATH	103	286
Total	154	1,213

[2011] produces a GENERALIZATION edge: the later work extends the relevant result from 2×2 matrices in SL_2 to higher-dimensional SL_n , preserving the form of the result while broadening the class of groups to which it applies. Details are in Appendix A.3.

Applying Reduce to each paper and Compare to candidate theorem pairs produces a TEG whose nodes are theorems and whose edges encode these evolutionary relationships.

Corpora. We construct three TEGs from 1,213 arXiv papers (Table 1). Two begin from recent datasets of mathematical research papers. We randomly sample 50 seed papers from OPENCONJECTURE [Brown, 2026] and 103 from ARXIVMATH [Dekoninck et al., 2026], then recursively follow their references to collect earlier work on which these papers build. At each step, we retain a candidate paper only when an LLM, given the titles and abstracts of the two papers, judges that their results are plausibly related at the theorem level. We recurse to depth 4 or until reaching 500 retained papers. This produces collections of 427 OPENCONJECTURE papers and 286 ARXIVMATH papers.

To evaluate the pipeline on an established line of mathematical development, we construct a TEG from Bridgeland [2007], a seminal paper introducing stability conditions on triangulated categories. We follow papers that cite it, yielding 500 papers that trace how this line of work has developed. Across the paper sets, the reduce step extracted main theorems from 1,128 papers, yielding 1,128 main-theorem nodes with supporting corollaries on which they depend. The compare step identified 285 evolutionary relations between main theorems. The graphs contain 1,255 edges in total, including those that connected supporting corollaries. Details are in Appendix A.

4 Experiments

Task and test set. Given a predecessor theorem, a model generates $k = 5$ conjectures that could plausibly follow from it. Our test set contains 50 predecessor theorems held out from the three TEGs: 26 from ARXIVMATH, 20 from BRIDGELAND, and 4 from OPENCONJECTURE. For each test predecessor, no theorem from any paper that cites it appears in the generator’s context under any condition, so the follow-up papers used for scoring are never visible to the generator. Each predecessor has published follow-up work that cites it.

Conditions. We compare five conditions that differ in the context provided to the generator. Zero-shot shows only the predecessor theorem. Citation graph additionally provides the titles, years, and abstracts of up to 30 papers cited by the predecessor’s source paper, testing whether TEG structure provides information beyond the bibliographic relationships already available in the literature. Two ablations separate the components of the TEG: reduce-only provides the reduced forms of neighboring theorems without evolutionary edges, and compare-only provides the full taxonomy without the neighboring theorem statements. TEG has both neighboring theorems and their typed evolutionary relations.

Models and evaluation. We run all conditions with gpt-5.4, gemini-2.5-pro, claude-opus-4-7, and deepseek-v3.2. Because conjecture quality is subjective, we evaluate generated conjectures retrospectively against published follow-up work. This serves as a proxy for research directions that mathematicians found interesting to pursue. For each predecessor, we retrieve papers that cite it and use an ensemble of gpt-5.4, claude-opus-4-7, and gemini-2.5-pro to judge each generated conjecture against each follow-up paper. Judges use a three-point scale: 3 for substantive similarity, 2 topical similarity, and 1 for no similarity. Scores are aggregated by the median across the three judges. Our primary metric is the fraction of

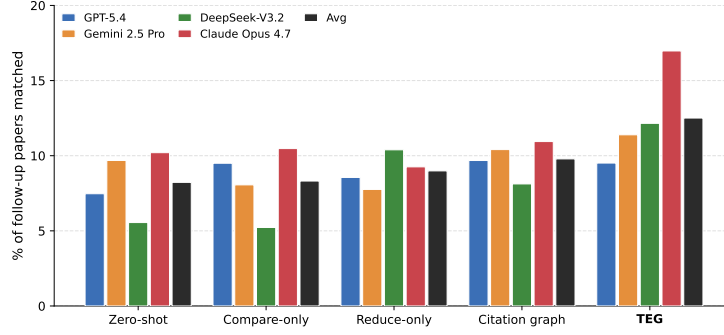


Figure 1: Fraction of published follow-up papers matched by at least one of $k = 5$ generated conjectures. TEG-augmented prompting performs best on average.

follow-up papers for which at least one of the $k = 5$ generated conjectures receives a 3. Prompts, model settings, costs, and human validation of the judges are in Appendix B.

Results. TEG-augmented prompting matches published follow-up work more often than the baselines (Figure 1). Averaged across the four generator models, TEG matches 12.5% of follow-up papers, compared with 9.8% for citation-graph context, 9.0% for reduce-only, 8.3% for compare-only, and 8.2% for zero-shot prompting. In relative terms, TEG matches 28% more follow-up papers than citation-graph context and 52% more than zero-shot prompting. Paired by predecessor and pooled across generators ($n = 179$), TEG improves over the citation graph by 2.9 percentage points (95% paired-bootstrap CI $[+0.7, +5.2]$; Wilcoxon $p = 0.002$). It also outperforms reduce-only (+3.5pp, $p = 2 \times 10^{-4}$), compare-only (+4.3pp, $p = 2 \times 10^{-5}$), and zero-shot prompting (+4.6pp, $p = 2 \times 10^{-7}$). Details in Appendix B.4.

The effect varies across generator models (Appendix Table 3). TEG improves over citation-graph context by 6.0pp for claude-opus-4-7 ($p = 0.002$) and 4.0pp for deepseek-v3.2 ($p = 0.03$), while the differences for gemini-2.5-pro (+1.0pp, $p = 0.31$) and gpt-5.4 (−0.2pp, $p = 0.80$) are not significant. We interpret this as a positive aggregate effect.

Generators may have seen follow-up papers during training. We repeat the analysis using only follow-up papers after each generator’s training cutoff (Appendix B.5). Averaged across generators, TEG matches 12.7% of post-cutoff follow-ups (11.4% pre-cutoff), compared with 10.7% for the citation graph (8.9% pre-cutoff). The TEG improvement over the citation graph is +2.6pp after the cutoff (95% CI $[-1.8, +7.2]$, $n = 131$) and +2.9pp before it.

Case study. We also asked two practicing mathematics professors to rate conjectures generated from papers in their own areas on 5-point scales for interestingness as a direction for future work and plausibility. TEG-augmented conjectures scored higher than citation-graph conjectures on both measures: 2.50 vs. 1.70 for interestingness and 3.40 vs. 2.90 for plausibility. The larger improvement in interestingness is consistent with the intended role of the TEG: helping models identify promising directions in which to extend prior results. The full protocol is in Appendix B.7.

5 Discussion

Future directions. Our results suggest that conjecture generation can be modeled as structured search over mathematical moves. A system could first choose how to extend a theorem, such as by generalizing it or weakening an assumption, and then instantiate that move as a conjecture. These moves could serve as actions in search or reinforcement learning, with rewards from signals such as counterexample resistance, provability, novelty, or expert preference. TEGs could also support human–AI workflows in which mathematicians request particular kinds of extensions and inspect the prior results motivating each candidate.

Limitations. We do not validate TEG construction against expert annotations at scale, so errors may propagate downstream. Our evaluation also uses similarity to published follow-up work as a

proxy for conjecture quality, which cannot capture good conjectures that were never pursued. TEGs also encode incremental development, such as generalizing a result, weakening a hypothesis, or transferring a technique. They therefore target conjectures that extend existing work, not conjectures that predict unexpected phenomena or connect distant areas. Finally, our experiments cover a limited set of mathematical corpora and use a fixed taxonomy.

Conclusion. We introduce the Theorem Evolution Graph (TEG), a representation of how mathematical theorems build on one another, and a Reduce–Compare pipeline for constructing TEGs from the literature. Across 1,213 papers, TEG context improves conjecture generation over citation-graph, ablation, and zero-shot baselines on average. More broadly, TEGs offer a way to use the structure of mathematical progress for the task of conjecture generation.

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A TEG Construction Details

A.1 Corpus collection

We construct the three paper collections using the Semantic Scholar Academic Graph API [Kinney et al., 2023]. For OPENCONJECTURE and ARXIVMATH, we begin from the sampled seed papers described in Section 3 and recursively follow their references. For the BRIDGELAND collection, we begin from Bridgeland [2007] and recursively follow papers that cite it. Each traversal stops at depth 4 or 500 retained papers.

Citation alone does not imply that two papers are related at the theorem level. We therefore apply an abstract-level filter before adding each candidate paper. Given the titles and abstracts of a cited paper and a citing paper, gpt-5.4 (temperature 0) predicts whether the citing paper’s mathematical results

plausibly depend on, generalize, extend, or apply results from the cited paper. We retain positive predictions with high or medium confidence and discard low-confidence positives. This filter is intended to remove citations used only for background, motivation, notation, or broad topical context.

The resulting collections contain 500 BRIDGELAND papers, 427 OPENCONJECTURE papers, and 286 ARXIVMATH papers, for 1,213 papers total. Reduce successfully extracts a main theorem from 1,128 papers.

A.2 Reduce implementation

For each paper, we provide its full source text to gpt-5.4 at temperature 0. We do not truncate or independently summarize the paper before extraction. The model is asked to identify the most central theorem and every definition, lemma, proposition, and corollary on which it depends, directly or transitively. The main theorem is returned as a complete mathematical statement, while supporting results are represented compactly. Each node additionally records its dependencies, producing a directed acyclic graph rooted at the main theorem.

We validate each returned graph before using it downstream. A valid reduction must contain a main theorem, every referenced dependency must correspond to another node in the same output, and the resulting dependency graph must be acyclic. Reduce outputs are cached by arXiv identifier and reused throughout the remaining pipeline.

Conceptually, the Reduce prompt asks the model to return the following structure:

```
MAIN THEOREM:
  name
  complete_statement
  assumptions
  proof_approach
  depends_on

SUPPORTING RESULTS:
[
  {
    id,
    type: definition | lemma | proposition | corollary,
    statement_or_summary,
    depends_on
  },
  ...
]
```

For papers with multiple major results, the model is instructed to select the theorem most central to the paper’s contribution.

A.3 Compare implementation

Compare first retrieves plausible theorem pairs and then classifies their relationship. We embed each reduced mathematical object with OpenAI’s text-embedding-3-small model (1,536 dimensions). The embedded representation concatenates the object’s name, mathematical statement, assumptions, and proof approach. Embeddings are indexed with HNSW, and candidate pairs below cosine similarity $\tau = 0.7$ are discarded.

For each retained pair, the earlier result is treated as the candidate predecessor and the later result as the candidate successor. Ordering is determined by arXiv publication date. We prompt gpt-5.4 at temperature 0 with both theorem representations and context from the later paper. The model first determines whether there is a substantive mathematical dependency or development. It then assigns one of the 13 relations in Table 2. Pairs that are merely topically similar, use a standard technique in common, or have no traceable mathematical dependence receive no edge.

The Compare response additionally includes a dependency strength (strong, moderate, weak, or none), an explanation of the assigned relationship, and textual evidence. We retain edges for which

Table 2: The 13 evolutionary relations used by Compare. The taxonomy is informed by prior accounts of mathematical discovery and development.

Relation	Description
GENERALIZATION	Extends an earlier result to a broader class of objects or assumptions.
SPECIALIZATION	Restricts an earlier result to a narrower case, often obtaining a sharper conclusion.
ASSUMPTION_WEAKENING	Establishes a related result under weaker hypotheses.
CONCLUSION_STRENGTHENING	Establishes a stronger conclusion under comparable hypotheses.
DOMAIN_EXTENSION	Applies an earlier result in a new mathematical setting.
PROOF_TECHNIQUE_TRANSPLANT	Reuses or adapts the proof method of an earlier theorem in a new context.
DIRECT_INVOCATION	Directly uses the earlier theorem as a lemma or step in proving the later result.
ANALOGY	Establishes a structurally parallel result in another mathematical setting.
ABSTRACTION_UNIFICATION	Places an earlier result within a more general framework that unifies multiple cases.
COUNTEREXAMPLE_TIGHTNESS	Establishes a limitation, sharpness result, or counterexample showing that an earlier bound cannot be improved.
REFORMULATION	Restates essentially the same mathematical result in a different representation or language.
LEMMA_INCORPORATION	Incorporates a supporting result from the earlier work into a later theorem or proof.
ALTERNATIVE_PROOF	Establishes the same or essentially equivalent result through a substantially different proof.

the model predicts a dependency with at least moderate strength. Across the three TEGs, Compare identifies 285 evolutionary relations between main theorems. Including lemma- and proposition-level dependencies, the graphs contain 1,255 edges.

Retrieval threshold. We considered cosine-similarity thresholds $\tau \in \{0.5, 0.6, 0.65, 0.7, 0.75\}$ on a small held-out set and selected $\tau = 0.7$. Lower thresholds substantially increased the number of LLM comparisons without a corresponding improvement in useful matches, while higher thresholds reduced retrieval recall.

B Experimental Details

B.1 Generation settings

We evaluate four generator models: gpt-5.4, gemini-2.5-pro, claude-opus-4-7, and deepseek-v3.2. Where supported, models are run at temperature 0.7 with a maximum output length of 16,000 tokens. gpt-5.4 is run using the provider’s default sampling configuration. Each generator produces $k = 5$ conjectures for every predecessor under each of the five experimental conditions.

The final test set contains 50 predecessor theorems: 26 from ARXIVMATH, 20 from BRIDGELAND, and 4 from OPENCONJECTURE. Each predecessor has at least one subsequent paper available for retrospective evaluation.

B.2 Conjecture-generation prompts

All five conditions use the same task framing and output format. The only difference is the context provided to the model. The common task instruction is:

You are an expert mathematician working in a specialized subfield.

You will be shown a predecessor theorem and, depending on the condition, additional context about related mathematical work.

Propose exactly k conjectures that could plausibly be the next mathematical result to follow from the predecessor. Each conjecture

should be a precisely stated open mathematical statement that another paper might pursue.

Return a JSON list of exactly k bare mathematical statements. Do not include titles, numbering, labels, or commentary.

The context differs by condition as follows.

Zero-shot. The prompt contains only the predecessor theorem. The model is instructed to reason from that theorem without additional literature context.

Citation graph. The prompt additionally contains bibliographic context from the predecessor’s source paper. Each included paper is represented by its title, year, and abstract.

Reduce-only. The prompt contains reduced representations of nearby theorems but removes the evolutionary relations connecting them. This condition tests whether the benefit of a TEG can be explained by providing additional, normalized mathematical statements without typed evolution information.

Compare-only. The prompt contains the evolutionary relation taxonomy but omits the contents of neighboring theorem statements. This condition tests whether the space of possible mathematical moves alone is sufficient to improve generation.

TEG. The full condition provides both the mathematical content of neighboring results and the evolutionary relations that connect them to related predecessor results. The prompt contains the predecessor’s local dependency graph and retrieved predecessor–successor examples from the TEG, including the typed relation associated with each pair. None of these theorems or retrieved examples comes from a paper that cites the test predecessor.

B.3 LLM evaluation

For each predecessor, we retrieve papers that subsequently cite its source paper. Each generated conjecture is compared against each such follow-up paper by three judge models: gpt-5.4, claude-opus-4-7, and gemini-2.5-pro. Each judge assigns a score from 1 to 3:

- 1: no meaningful relationship beyond, at most, a broad shared area;
- 2: topical similarity, but the paper pursues a substantively different mathematical question;
- 3: substantive alignment, meaning that the paper addresses what the conjecture proposes or a clear variant of it.

Judges are explicitly instructed to prefer 2 when uncertain between 2 and 3. Scores are aggregated by taking the median of the three judges for each conjecture–paper pair, which is equivalent to a majority decision for the integer-valued three-judge ensemble.

The core judge prompt is:

You are an expert research mathematician.

You will see:

- (1) a predecessor theorem,
- (2) k candidate conjectures proposed as follow-up directions, and
- (3) n published papers that subsequently cite the predecessor.

For every (conjecture, citing-paper) pair, judge how closely the conjecture aligns with the research direction actually taken by the published paper.

Score:

1 = no meaningful relationship

Table 3: TEG minus citation-graph performance by generator.

Generator	Difference (pp)	p
gpt-5.4	-0.2	0.80
gemini-2.5-pro	+1.0	0.31
deepseek-v3.2	+4.0	0.03
claude-opus-4-7	+6.0	0.002

Table 4: Primary metric (% of follow-up papers matched), averaged over generators, by follow-up date relative to each generator’s training cutoff. n is the number of paired (predecessor, generator) observations.

Follow-ups	n	Zero-shot	Compare-only	Reduce-only	Citation	TEG
All	179	8.2	8.3	9.0	9.8	12.5
Pre-cutoff	171	6.3	8.2	7.4	8.9	11.4
Post-cutoff	131	8.2	10.5	10.7	10.7	12.7

2 = same area, but substantively different direction

3 = strong alignment; the paper substantively addresses what the conjecture proposes, or a clear variant of it

When uncertain between 2 and 3, use 2.

Return one score for each of the $k \times n$ pairs and no additional commentary.

Our primary outcome is the fraction of follow-up papers for which at least one of the five conjectures receives an aggregated score of 3.

B.4 Statistical analysis

Not every (predecessor, generator) pair yields a parseable generation; the four generators had 179 paired observations in total. For each generator and predecessor, we compute the fraction of follow-up papers matched under each condition. Comparisons are paired by predecessor. For pooled analyses, we treat each available (predecessor, generator) pair as one paired observation and pool across the four generator models.

We compute 95% confidence intervals for the mean paired difference using 10,000 bootstrap resamples of these paired units. We additionally use a two-sided Wilcoxon signed-rank test for the null hypothesis of no condition-level difference.

In the final experiment, TEG improves over citation-graph context by 2.9 percentage points (95% paired-bootstrap CI $[+0.7, +5.2]$, $p = 0.002$). The corresponding differences are +3.5pp over reduce-only ($p = 2 \times 10^{-4}$), +4.3pp over compare-only ($p = 2 \times 10^{-5}$), and +4.6pp over zero-shot prompting ($p = 2 \times 10^{-7}$).

B.5 Training-cutoff analysis

We use the training cutoffs reported by each provider: January 2025 (gemini-2.5-pro), July 2025 (deepseek-v3.2), August 2025 (gpt-5.4), and January 2026 (claude-opus-4-7). We date each follow-up paper by its first arXiv posting. For the 23% of follow-ups without an arXiv identifier, we use the Semantic Scholar publication year and count the paper as post-cutoff only if that year is later than the cutoff year. For each generator, we keep only follow-ups posted after its cutoff and recompute the primary metric on the predecessors used in the main analysis, dropping predecessors with no post-cutoff follow-ups. Generations and judgments are unchanged; only the set of follow-ups being scored changes.

The post-cutoff subset is smaller (131 paired observations; 628 follow-ups), so intervals are wide. Effects also vary by generator. For claude-opus-4-7, which has the latest cutoff, the TEG advantage

Table 5: Agreement between the three-model LLM judge ensemble and a practicing mathematician.

Metric	Value ($n = 14$)
Exact agreement	50.0% (7/14)
Within ± 1	92.9% (13/14)
Mean absolute error	0.57
Precision on 3	0.57 (4/7)
Recall on 3	1.00 (4/4)

over the citation graph shrinks from +10.0pp before the cutoff to +1.9pp after it (71 post-cutoff follow-ups). For gpt-5.4 it grows from -2.2pp to +1.4pp.

B.6 Validation of the LLM judges

To evaluate whether the LLM ensemble provides a reasonable approximation to expert mathematical judgment, we randomly sample 15 (conjecture, follow-up paper) pairs spanning all four generator models and all five context conditions. A practicing mathematician independently scores each pair using the same three-point rubric. One generated conjecture is ill-formed and cannot be adjudicated, leaving 14 scored pairs.

Table 5 reports agreement. The ensemble exactly matches the expert on 7 of 14 pairs and is within one scale point on 13 of 14. On the primary distinction between substantive alignment (3) and lower scores, the ensemble identifies all four pairs that the expert scores as 3, but also labels three additional pairs as 3. Thus, this small validation suggests high recall but more limited precision for the positive class.

B.7 Expert case study

Two practicing mathematics professors evaluate conjectures generated from papers in their own research areas. Each rater receives one predecessor theorem and 10 candidate conjectures: five produced with citation-graph context and five produced with TEG context. Conjectures are presented in random order, and condition labels are hidden.

For each conjecture, raters answer two questions on 5-point scales:

- **Interestingness:** “How interesting is this conjecture as a direction for future work in your field?”
- **Plausibility:** “How plausibly true does this conjecture seem to you?”

Aggregated across the two raters, TEG-augmented conjectures receive a mean interestingness score of 2.50, compared with 1.70 for citation-graph conjectures. Mean plausibility is 3.40 for TEG and 2.90 for citation-graph.

The raters were recruited through the authors’ professional contacts in a university mathematics department. Participation was voluntary, and raters received no monetary compensation. Individual ratings are reported only in aggregate.

B.8 Computational budget

TEG construction requires approximately one Reduce call per source paper, approximately 47,300 Compare calls for candidate theorem pairs surfaced by embedding retrieval, and approximately 6,500 abstract-level filtering calls during corpus expansion. The original construction run used approximately 350M input tokens and 55M output tokens, with an estimated API cost of approximately \$2,500.

The conjecture-generation evaluation consists of 1,000 generation calls (50 predecessors $\times$ 4 generator models $\times$ 5 conditions) and approximately 3,000 judge calls. The original experiment consumed approximately 200M tokens at an estimated API cost of approximately \$1,200. Embedding retrieval was run on a single workstation using `text-embedding-3-small` and an HNSW index.